

Unit 12. Principal Components and Low-Rank Approximation

1. Centered data and covariance

Let $D \in \mathbb{R}^{n \times p}$ contain $n \geq 2$ observations as rows and p variables as columns. Put

$$\bar{d} = \frac{1}{n} D^\top \mathbf{1} \in \mathbb{R}^p, \quad C = I_n - \frac{1}{n} \mathbf{1} \mathbf{1}^\top, \quad Z = CD = D - \mathbf{1} \bar{d}^\top.$$

Thus row i of Z is the deviation of observation i from the mean row. Since $\mathbf{1}^\top \mathbf{1} = n$, $C^\top = C$, $C^2 = C$, $C\mathbf{1} = 0$ and $\mathbf{1}^\top Z = 0$. The sample covariance matrix is

$$S = \frac{1}{n-1} Z^\top Z \in \mathbb{R}^{p \times p}.$$

Its entry s_{jk} is the sample covariance between columns j, k . It is symmetric positive semidefinite because $a^\top S a = \|Za\|^2 / (n-1) \geq 0$. Moreover $\ker S = \ker Z$, so

$$\text{rank } S = \text{rank } Z \leq \min(p, n-1), \quad \text{tr } S = \frac{\|Z\|_F^2}{n-1} = \sum_{j=1}^p s_{jj}.$$

The rank bound uses $\text{col}(Z) \subseteq \mathbf{1}^\perp$. The total variance is the sum of the individual variable variances. For $a \in \mathbb{R}^p$, the column Za contains the centered values of the linear combination with coefficients a . Its mean is zero and its sample variance is $a^\top S a$. A norm constraint on a is essential: replacing a by ta multiplies the variance by t^2 .

Example 12.1 (A complete centered covariance calculation). Take

$$D = \begin{pmatrix} 12 & 21 \\ 11 & 22 \\ 9 & 18 \\ 8 & 19 \end{pmatrix}.$$

The mean is $(10, 20)^\top$, giving

$$Z = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{pmatrix}, \quad Z^\top Z = \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}, \quad S = \frac{1}{3} \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}.$$

For $a = e_1$, the score variance is $10/3$; for $a = (1, 1)^\top / \sqrt{2}$, it is $(10 + 8 + 8 + 10)/6 = 6$. The unit-length direction changes which linear combination is being measured while fixing the coefficient scale.

2. Principal directions and scores

Write the spectral decomposition

$$S = V \Lambda V^\top, \quad V = (v_1 \cdots v_p), \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p), \quad \lambda_1 \geq \cdots \geq \lambda_p \geq 0.$$

The **principal directions**, or loading vectors in these notes, are the unit columns v_j . The **score matrix** is $T = ZV \in \mathbb{R}^{n \times p}$, with score column $t_j = Zv_j$. The meaning of “loading” is fixed here as the unit eigenvector, not an eigenvector multiplied by a square root of its eigenvalue.

Theorem 12.1 (Successive maximum variance). Among unit a , the maximum of $a^\top Sa$ is λ_1 . After choosing orthonormal eigenvectors $v_1, \dots, v_{k-1}$ in decreasing eigenvalue order, the maximum under the additional restrictions $a^\top v_i = 0$ for $i < k$ is λ_k , attained at v_k . The score columns have sample covariance $T^\top T / (n - 1) = \Lambda$.

Proof. Expand $a = \sum_i c_i v_i$. The unit constraint becomes $\sum_i c_i^2 = 1$, and $a^\top Sa = \sum_i \lambda_i c_i^2$. It is at most λ_1 , with equality precisely when $c_i = 0$ for every $\lambda_i < \lambda_1$. The extra orthogonality restrictions set $c_1 = \dots = c_{k-1} = 0$. The remaining weighted average is at most λ_k , attained by putting $c_k = 1$ and all other coefficients zero. Its equality vectors are the unit vectors in the still-available part of the λ_k eigenspace. Finally, $T^\top T / (n - 1) = V^\top S V = \Lambda$. $\square$

Zero covariance between score columns follows from this matrix identity. Statistical independence is a separate assertion and need not hold. When $\text{tr } S > 0$, the variance proportion of component j is $\lambda_j / \text{tr } S$, and the first k components explain $\sum_{j=1}^k \lambda_j / \text{tr } S$ of total variance. If $S = 0$, all observations equal their mean and these proportions have zero denominator; no direction has positive variance.

Example 12.2 (Scores, reconstruction, and a residual). For Example 12.1,

$$\lambda_1 = 6, \quad \lambda_2 = \frac{2}{3}, \quad v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The scores are

$$T = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & 1 \\ 3 & -1 \\ -3 & 1 \\ -3 & -1 \end{pmatrix}.$$

Their squared lengths are 18 and 2, and their dot product is zero. Dividing by $n - 1 = 3$ recovers variances 6 and $2/3$. The first direction explains $6/(6 + 2/3) = 9/10$ of total variance. Keeping only its score reconstructs

$$Z_1 = t_1 v_1^\top = \frac{1}{2} \begin{pmatrix} 3 & 3 \\ 3 & 3 \\ -3 & -3 \\ -3 & -3 \end{pmatrix}, \quad D_1 = \mathbf{1} \bar{d}^\top + Z_1.$$

For example the first reconstructed observation is (11.5, 21.5). The residual is

$$Z - Z_1 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 1 & -1 \\ -1 & 1 \end{pmatrix},$$

whose squared Frobenius norm is $8/4 = 2$. The lost variance is $2/(n - 1) = 2/3$, exactly the discarded eigenvalue.

3. Variance captured by a subspace

Let $Q \in \mathbb{R}^{p \times k}$ have orthonormal columns. Projection of each centered row onto $\text{col}(Q)$ gives $ZQQ^\top$. Its score coordinates in this basis are ZQ , and

$$\frac{\|ZQ\|_F^2}{n - 1} = \text{tr}(Q^\top S Q).$$

Replacing Q by QR for an orthogonal $k \times k$ matrix R rotates the score coordinates but leaves $QQ^\top$ and the reconstruction unchanged.

Theorem 12.2 (Maximum captured variance). For any symmetric $p \times p$ matrix A with eigenvalues in decreasing order and any $1 \leq k \leq p$,

$$\max_{Q^\top Q = I_k} \operatorname{tr}(Q^\top A Q) = \sum_{i=1}^k \lambda_i(A), \quad \min_{Q^\top Q = I_k} \operatorname{tr}(Q^\top A Q) = \sum_{i=p-k+1}^p \lambda_i(A).$$

For the maximum, an optimal subspace contains every eigenspace with eigenvalue strictly greater than λ_k and is contained in the sum of the eigenspaces with eigenvalue at least λ_k .

Proof. Let $A = V\Lambda V^\top$ and $P = QQ^\top$. Put $w_i = v_i^\top P v_i$. Since P is an orthogonal projector, $w_i = \|P v_i\|^2 \in [0, 1]$ and $\sum_i w_i = \operatorname{tr} P = k$. Cyclic trace gives $\operatorname{tr}(Q^\top A Q) = \sum_i \lambda_i w_i$. Set $\alpha = \lambda_k$. Then

$$\sum_{i=1}^k \lambda_i - \sum_{i=1}^p \lambda_i w_i = \sum_{i \leq k} (\lambda_i - \alpha)(1 - w_i) + \sum_{i > k} (\alpha - \lambda_i) w_i \geq 0.$$

Here the terms in α cancel because $\sum_i w_i = k$. The upper bound is attained by $Q = (v_1 \cdots v_k)$. Equality forces $w_i = 1$ when $\lambda_i > \alpha$ and $w_i = 0$ when $\lambda_i < \alpha$. The identities $w_i = 1 \iff P v_i = v_i$ and $w_i = 0 \iff P v_i = 0$ give the stated subspace characterization, including ties. Conversely those conditions give equality. Apply the upper-bound result to $-A$ to obtain the minimum, attained by the bottom k eigenvectors. $\square$

If $k < p$ and $\lambda_k > \lambda_{k+1}$, the optimal k -dimensional subspace is unique, although its orthonormal basis is not. If the cutoff lies inside a repeated eigenspace, several optimal subspaces may exist. Changing the sign of a loading also changes the sign of its score, leaving their product $t_j v_j^\top$ unchanged.

4. Best low-rank reconstruction

The Frobenius norm satisfies $\|A\|_F^2 = \sum_{i,j} a_{ij}^2 = \operatorname{tr}(A^\top A)$ and is unchanged by orthogonal multiplication on either side. If P is an orthogonal projector, rowwise Pythagoras gives

$$\|Z\|_F^2 = \|ZP\|_F^2 + \|Z(I - P)\|_F^2.$$

For a fixed P , ZP is the closest matrix to Z whose rows lie in $\operatorname{col}(P)$.

Theorem 12.3 (Best approximation of prescribed rank). Let $Z \in \mathbb{R}^{n \times p}$ have positive singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$ and SVD $Z = \sum_{i=1}^r \sigma_i u_i v_i^\top$. For $0 \leq k \leq r$, put $Z_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$ (the empty sum is zero). Then

$$\min_{\operatorname{rank} B \leq k} \|Z - B\|_F^2 = \|Z - Z_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2.$$

For $k \geq r$ the minimum is zero, attained by $B = Z$.

Proof. Let B have row space of dimension $l \leq k$ and let P project onto that space. Since $B = BP$,

$$Z - B = Z(I - P) + (ZP - B).$$

The rows of the two summands belong to perpendicular subspaces, so

$$\|Z - B\|_F^2 = \|Z(I - P)\|_F^2 + \|ZP - B\|_F^2 \geq \|Z\|_F^2 - \operatorname{tr}(PZ^\top Z).$$

The trace theorem applied to $Z^\top Z$ gives $\operatorname{tr}(PZ^\top Z) \leq \sum_{i=1}^l \sigma_i^2 \leq \sum_{i=1}^k \sigma_i^2$, with zero eigenvalues appended as needed. Thus every admissible B has error at least the displayed tail sum. Taking $P = \sum_{i=1}^k v_i v_i^\top$ gives $ZP = Z_k$, of rank at most k . The matrices $u_i v_i^\top$ are orthonormal in the Frobenius product, because their pairwise inner products are $(u_i^\top u_j)(v_i^\top v_j) = \delta_{ij}$. Therefore $\|Z - Z_k\|_F^2 = \sum_{i > k} \sigma_i^2$, attaining the lower bound. The cases $k = 0$ and $k \geq r$ follow directly as well. $\square$

For centered data, $\sigma_i^2 = (n-1)\lambda_i(S)$. The theorem therefore proves both the global minimum among *all* rank- k matrices and the PCA residual formula

$$\|Z - Z_k\|_F^2 = (n-1) \sum_{i>k} \lambda_i(S).$$

It does not restrict competitors in advance to matrices of the form $ZQQ^\top$; the proof shows why an optimal competitor can be chosen in that form. The centered low-rank fit to the original observations is $D_k = \mathbf{1}\bar{d}^\top + Z_k$.

For an arbitrary affine subspace $a + W$ of dimension at most k , with orthogonal projector P onto W , its total squared distance from the observations is

$$\sum_{i=1}^n \|(I-P)(d_i - a)\|^2 = \|Z(I-P)\|_F^2 + n \|(I-P)(\bar{d} - a)\|^2.$$

The cross term vanishes because the centered rows sum to zero. For each direction space W , the minimum is achieved exactly when $(I-P)a = (I-P)\bar{d}$, meaning the affine subspace contains the mean. Optimizing over W then gives the PCA subspace through the mean.

5. SVD, sample-space computation, and scaling

With $Z = U_r \Sigma_r V_r^\top$,

$$S = V_r \frac{\Sigma_r^2}{n-1} V_r^\top, \quad G = \frac{1}{n-1} Z Z^\top = U_r \frac{\Sigma_r^2}{n-1} U_r^\top.$$

The covariance matrix S is $p \times p$ and concerns variable directions; the Gram matrix G is $n \times n$ and concerns observation coordinates. They share their positive eigenvalues, with potentially different counts of zero eigenvalues. For a unit u with $Gu = \lambda u$, $\lambda > 0$,

$$v = \frac{Z^\top u}{\sqrt{(n-1)\lambda}}, \quad Sv = \lambda v, \quad \|v\| = 1, \quad Zv = \sqrt{(n-1)\lambda} u.$$

The norm identity follows from $\|Z^\top u\|^2 = u^\top Z Z^\top u = (n-1)\lambda$. This provides loadings and scores from the smaller matrix when p is large compared with n . The formula divides by $\sqrt{\lambda}$ and therefore applies only to positive eigenvalues; null directions are obtained separately from $\ker Z$.

For the running example, $\sigma_1 = \sqrt{18} = 3\sqrt{2}$ and $\sigma_2 = \sqrt{2}$; the left singular columns are

$$u_1 = \frac{1}{2}(1, 1, -1, -1)^\top, \quad u_2 = \frac{1}{2}(1, -1, 1, -1)^\top.$$

Thus G has eigenvalues 6, 2/3, 0, 0 while S has 6, 2/3.

Suppose all variable variances are positive and put $D_s = \text{diag}(\sqrt{s_{11}}, \dots, \sqrt{s_{pp}})$. Standardizing each column gives $Z_s = Z D_s^{-1}$ and correlation matrix

$$R = \frac{1}{n-1} Z_s^\top Z_s = D_s^{-1} S D_s^{-1}.$$

This is a congruence, not an orthogonal similarity. It changes the optimization problem: correlation PCA measures variance after each variable is rescaled to variance one. If a column is constant, its variance is zero and division by that standard deviation is undefined. Such a column contributes no centered variation and is removed from the standardized calculation.

Example 12.3 (Scaling can change the selected subspace). If $S = \text{diag}(9, 1)$, covariance PCA uniquely selects the first coordinate as its leading one-dimensional subspace and explains 9/10 of total variance. Standardization gives $R = I_2$. Every unit direction is then optimal and explains 1/2 of the standardized total variance. Equalized units have changed both the optimum and its uniqueness. For the running two-variable data, the two variances are equal, so standardization multiplies all of S by the same scalar; the directions and 9/10 proportion remain unchanged.

6. Min–max and compression: a variational extension

These formulas extend the variance arguments to interior eigenvalues and to matrices obtained by selecting or projecting variables. For a symmetric $A \in \mathbb{R}^{p \times p}$ and $x \neq 0$, write $\rho_A(x) = x^\top A x / (x^\top x)$.

Theorem 12.4 (Min–max formulas). For eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$,

$$\lambda_k = \max_{\dim W=k} \min_{0 \neq x \in W} \rho_A(x) = \min_{\dim L=p-k+1} \max_{0 \neq x \in L} \rho_A(x).$$

Proof. For any k -dimensional W , the dimension formula implies $W \cap \text{span}(v_k, \dots, v_p) \neq \{0\}$, since the two dimensions sum to $p+1$. On this intersection a nonzero vector has Rayleigh quotient at most λ_k , so the inner minimum on W is at most λ_k . For $W = \text{span}(v_1, \dots, v_k)$, every nonzero Rayleigh quotient is at least λ_k and equality occurs at v_k . This proves the first formula, including attainment. For any $(p-k+1)$ -dimensional L , its intersection with $\text{span}(v_1, \dots, v_k)$ is nonzero. Thus its inner maximum is at least λ_k . Choosing $L = \text{span}(v_k, \dots, v_p)$ attains this lower bound. $\square$

Theorem 12.5 (Separation under compression). If $Q \in \mathbb{R}^{p \times r}$ has orthonormal columns and $\mu_1 \geq \dots \geq \mu_r$ are the eigenvalues of $Q^\top A Q$, then

$$\lambda_{p-r+j}(A) \leq \mu_j \leq \lambda_j(A) \quad (1 \leq j \leq r).$$

Proof. The map $y \mapsto Qy$ preserves lengths and dimensions, and $\rho_{Q^\top A Q}(y) = \rho_A(Qy)$. The max–min formula for μ_j ranges over j -dimensional subspaces contained in $\text{col}(Q)$, a subset of the competitors for $\lambda_j(A)$. Thus $\mu_j \leq \lambda_j(A)$. The min–max formula for μ_j ranges over $(r-j+1)$ -dimensional subspaces contained in that column space. Restricting the competitors can only increase the minimum, so it is at least $\lambda_{p-(r-j+1)+1}(A) = \lambda_{p-r+j}(A)$. $\square$

Taking columns of Q to be coordinate vectors yields the interlacing bounds for any principal submatrix. Taking A to be a covariance matrix shows how the eigenvalues of a selected-variable covariance are bounded by those of the full covariance. Individual variances need not be eigenvalues, even though their sum always equals the eigenvalue sum.

Exercises and solutions

Centering and spectral decompositions

Exercise 12.1 (Centering and eigenvalue dispersion). Let $C = I_n - \mathbf{1}\mathbf{1}^\top/n$. Prove that C is symmetric idempotent, that $(Cx)_j = x_j - \bar{x}$, and that $C\mathbf{1} = 0$. For symmetric positive semidefinite A , prove $\text{tr}(A^2) - (\text{tr } A)^2/n \geq 0$.

Solution. The matrix $J = \mathbf{1}\mathbf{1}^\top/n$ is symmetric and $J^2 = \mathbf{1}(\mathbf{1}^\top\mathbf{1})\mathbf{1}^\top/n^2 = J$. Thus $C^\top = C$ and $C^2 = I - 2J + J^2 = C$. Since $Jx = \bar{x}\mathbf{1}$, the coordinate claim follows, and putting $x = \mathbf{1}$ gives $C\mathbf{1} = 0$. If $\lambda = (\lambda_1, \dots, \lambda_n)^\top$ lists the eigenvalues of A , spectral diagonalization gives

$$\text{tr}(A^2) - \frac{(\text{tr } A)^2}{n} = \sum_i (\lambda_i - \bar{\lambda})^2 = \lambda^\top C \lambda \geq 0.$$

Equality holds exactly when all eigenvalues coincide, equivalently $A = \bar{\lambda}I$. The same inequality in fact holds for every real symmetric A .

Exercise 12.2 (A resolution of the identity). Let $P_1, \dots, P_m$ be symmetric idempotent $n \times n$ matrices with $\sum_i P_i = I$ and ranks r_i . Prove $\sum_i r_i = n$, $P_i P_j = 0$ for $i \neq j$, and that $P_i + P_j$ is symmetric idempotent.

Solution. Since the trace of each projector equals its rank, $\sum_i r_i = \text{tr}(\sum_i P_i) = n$. For $x \in \text{col}(P_j)$, $P_j x = x$ and

$$\|x\|^2 = x^\top \left(\sum_i P_i \right) x = \|x\|^2 + \sum_{i \neq j} \|P_i x\|^2.$$

Every summand in the remaining sum is nonnegative, so $P_i x = 0$ for $i \neq j$. Apply this to $x = P_j y$ for arbitrary y to get $P_i P_j = 0$. Transposition also gives $P_j P_i = 0$. Thus $(P_i + P_j)^2 = P_i + P_j$, and symmetry is inherited from the two summands. This proves all assertions without a separate rank-inequality criterion for idempotence.

Exercise 12.3 (Weighted sums of perpendicular projectors). Under the preceding hypotheses, let $A = \sum_i a_i P_i$. Find its eigenvalues with multiplicities, its determinant, and its inverse when every $a_i \neq 0$.

Solution. On $\text{col}(P_i)$ the matrix acts as $a_i I$, since $AP_i = a_i P_i$. The ranges are perpendicular and their direct sum is $\mathbb{R}^n$, as $x = \sum_i P_i x$ for every x . Choose an orthonormal basis in each range and concatenate them. The result diagonalizes A with a_i repeated r_i times; if several a_i coincide their multiplicities add. Ranges of rank zero contribute no eigenvectors. Thus $\det A = \prod_i a_i^{r_i}$, with zero-rank factors omitted. If all $a_i \neq 0$, then

$$A \left(\sum_i a_i^{-1} P_i \right) = \sum_i P_i = I,$$

and the reversed product is also I , so $A^{-1} = \sum_i a_i^{-1} P_i$.

Exercise 12.4 (Equicorrelation). For $n \geq 2$, let A have diagonal entries 1 and off-diagonal entries t . Find its eigenvalues/eigenvectors, determinant, inverse when it exists, and the exact range of t for positive definiteness.

Solution. With $J = \mathbf{1}\mathbf{1}^\top/n$ and $C = I - J$,

$$A = (1 - t)I + t\mathbf{1}\mathbf{1}^\top = [1 + (n - 1)t]J + (1 - t)C.$$

Its eigenvalue on $\mathbb{R}\mathbf{1}$ is $a = 1 + (n - 1)t$; on $\mathbf{1}^\perp$ it is $b = 1 - t$, of multiplicity $n - 1$. For $t \neq 0$ these two eigenvalues are distinct, so their eigenspaces are exactly those two subspaces. For $t = 0$, $A = I$ and every nonzero vector is an eigenvector. An orthonormal eigenbasis is always $\mathbf{1}/\sqrt{n}$ followed by any orthonormal basis of $\mathbf{1}^\perp$. The determinant is ab^{n-1} . It is invertible exactly when $t \notin \{1, -1/(n - 1)\}$, with $A^{-1} = a^{-1}J + b^{-1}C$. Positive definiteness requires both $a, b > 0$, hence $-1/(n - 1) < t < 1$. At the two endpoints it is positive semidefinite and singular, with the zero eigenspace identified by the same decomposition.

Exercise 12.5 (Pairwise differences). Find the matrix of the quadratic form $\sum_{i,j=1}^n (x_i - x_j)^2$ and all its eigenvalues and eigenspaces.

Solution. Expansion gives

$$\sum_{i,j} (x_i - x_j)^2 = 2n \sum_i x_i^2 - 2 \left(\sum_i x_i \right)^2 = x^\top (2nI - 2\mathbf{1}\mathbf{1}^\top) x = 2nx^\top Cx.$$

For $n > 1$, the eigenvalue on $\mathbb{R}\mathbf{1}$ is 0 and on $\mathbf{1}^\perp$ it is $2n$, with multiplicities 1 and $n - 1$. Thus an orthonormal eigenbasis is the normalized constant vector and any orthonormal basis of its perpendicular complement. For $n = 1$, the form is identically zero and its only eigenvalue is 0. The identity also shows that total pairwise squared differences equal $2n \sum_i (x_i - \bar{x})^2$.

Variational formulas and compression

In the next ten exercises, A is real symmetric of order n , $\lambda_1 \geq \dots \geq \lambda_n$ are its eigenvalues, $s_1, \dots, s_n$ is a corresponding orthonormal eigenbasis, and every Rayleigh quotient is evaluated only at a nonzero vector.

Exercise 12.6 (Excluding selected eigenvectors). Let $R_j = (s_1 \ \dots \ s_j)$ and $T_j = (s_j \ \dots \ s_n)$. Prove

$$\lambda_k = \max_{R_{k-1}^\top x = 0} \rho_A(x) = \min_{T_{k+1}^\top x = 0} \rho_A(x).$$

An empty matrix of constraints imposes no restriction.

Solution. Write $x = \sum_i c_i s_i$. In the first feasible set, $c_1 = \dots = c_{k-1} = 0$, so $\rho_A(x) = \sum_{i=k}^n \lambda_i c_i^2 / \sum_{i=k}^n c_i^2 \leq \lambda_k$. Equality occurs at $x = s_k$. In the second feasible set $c_{k+1} = \dots = c_n = 0$, so the analogous weighted average of $\lambda_1, \dots, \lambda_k$ is at least λ_k , again attained at s_k . For $k = 1$ in the maximum and $k = n$ in the minimum, the corresponding coefficient restriction is empty, giving the usual extreme Rayleigh bounds.

Exercise 12.7 (Arbitrary linear restrictions). For every $B \in \mathbb{R}^{n \times (k-1)}$ and $C \in \mathbb{R}^{n \times (n-k)}$, prove

$$\min_{C^\top x=0} \rho_A(x) \leq \lambda_k \leq \max_{B^\top x=0} \rho_A(x).$$

Do not assume that the columns of B or C are independent.

Solution. Let $R = (s_1 \cdots s_k)$. The matrix $B^\top R$ has k columns and at most $k-1$ rows, so some nonzero u satisfies $B^\top Ru = 0$. Then $x = Ru \neq 0$ is feasible and $\rho_A(x) = \sum_{i=1}^k \lambda_i u_i^2 / \sum_i u_i^2 \geq \lambda_k$. This proves the right inequality. For the left, take $T = (s_k \cdots s_n)$, whose $n-k+1$ columns exceed the row count of $C^\top T$. Choose $v \neq 0$ in its kernel. Then $Tv \neq 0$ is feasible and has Rayleigh quotient at most λ_k . Dependence among the constraint columns only increases these kernels, so it causes no problem.

Exercise 12.8 (Fischer's min-max formulas). Prove

$$\lambda_k = \min_{B^\top B = I_{k-1}} \max_{B^\top x=0} \rho_A(x) = \max_{C^\top C = I_{n-k}} \min_{C^\top x=0} \rho_A(x),$$

where B and C have the dimensions in the preceding problem.

Solution. For every admissible B , the preceding problem makes the inner maximum at least λ_k . Choosing $B = (s_1 \cdots s_{k-1})$ makes it exactly λ_k by Exercise 12.6. Thus the outer minimum is λ_k and is attained. For every admissible C , the inner minimum is at most λ_k . Choosing $C = (s_{k+1} \cdots s_n)$ attains equality, so the outer maximum is λ_k . The empty-column cases work by the stated constraint convention.

Exercise 12.9 (Eigenvalue monotonicity). If B is positive semidefinite, prove $\lambda_k(A+B) \geq \lambda_k(A)$ for every k . Show the inequalities are strict when B is positive definite.

Solution. For every nonzero x , $\rho_{A+B}(x) = \rho_A(x) + \rho_B(x) \geq \rho_A(x) + \lambda_n(B)$. Apply the maximum over k -dimensional subspaces of the minimum Rayleigh quotient on each subspace to obtain $\lambda_k(A+B) \geq \lambda_k(A) + \lambda_n(B)$. For $B \succeq 0$, $\lambda_n(B) \geq 0$; for $B \succ 0$, it is strictly positive. Thus positive definiteness gives strict increase for every ordered eigenvalue, even if A itself is indefinite.

Exercise 12.10 (* Poincaré separation). Let $G \in \mathbb{R}^{n \times r}$ satisfy $G^\top G = I_r$. If $\mu_1 \geq \cdots \geq \mu_r$ are the eigenvalues of $G^\top A G$, prove $\lambda_{n-r+k} \leq \mu_k \leq \lambda_k$.

Solution. The map $y \mapsto Gy$ preserves lengths, dimensions and Rayleigh quotients. By max-min, μ_k is the maximum, over k -dimensional subspaces contained in $\text{col}(G)$, of the minimum of ρ_A on that subspace. Removing the restriction that the subspace lie in $\text{col}(G)$ can only increase the maximum, giving $\mu_k \leq \lambda_k$. By min-max, μ_k is the minimum of the maximum over $(r-k+1)$ -dimensional subspaces in $\text{col}(G)$. Allowing all subspaces of that dimension can only decrease the minimum, giving the bound $\mu_k \geq \lambda_{n-r+k}$. Both arguments include $r = n$ and the endpoint eigenvalues.

Exercise 12.11 (Compression by an orthogonal projector). Let $P = P^\top = P^2$ have rank $r \geq 1$. Write the eigenvalues of PAP , apart from $n-r$ zero eigenvalues, as $\mu_1 \geq \cdots \geq \mu_r$. Prove $\lambda_{n-r+k} \leq \mu_k \leq \lambda_k$.

Solution. Choose an orthonormal basis G of the range of P and complete it by F to an orthogonal matrix. Then $P = GG^\top$ and

$$(G \ F)^\top P A P (G \ F) = \begin{pmatrix} G^\top A G & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus the r listed eigenvalues are those of $G^\top A G$, including any zeros among those r values. Apply the preceding separation theorem. The wording does not require PAP to have rank exactly r .

Exercise 12.12 (Principal submatrices). For any $r \times r$ principal submatrix A_J of A , prove $\lambda_{n-r+k}(A) \leq \lambda_k(A_J) \leq \lambda_k(A)$. Write the full interlacing chain when $r = n - 1$.

Solution. Let G consist of the coordinate columns indexed by J . Then $G^\top G = I_r$ and $G^\top A G = A_J$, so separation applies. For $r = n - 1$, putting $\mu_k = \lambda_k(A_J)$ gives $\lambda_{k+1} \leq \mu_k \leq \lambda_k$, or in full order,

$$\lambda_n \leq \mu_{n-1} \leq \lambda_{n-1} \leq \mu_{n-2} \leq \cdots \leq \mu_1 \leq \lambda_1.$$

The selected indices need not be the first r indices.

Exercise 12.13 (Trace bounds and attaining subspaces). Prove that the maximum and minimum of $\text{tr}(Q^\top A Q)$ over $Q^\top Q = I_r$ are respectively $\sum_{k=1}^r \lambda_k$ and $\sum_{k=1}^r \lambda_{n-r+k}$. Deduce the same bounds for the trace of the leading $r \times r$ principal submatrix.

Solution. Let $\mu_1, \dots, \mu_r$ be the eigenvalues of $Q^\top A Q$. Separation gives $\lambda_{n-r+k} \leq \mu_k \leq \lambda_k$. Summing yields the two bounds since $\text{tr}(Q^\top A Q) = \sum_k \mu_k$. The upper bound is attained by $Q = (s_1 \cdots s_r)$; the lower by $Q = (s_{n-r+1} \cdots s_n)$. Taking $Q = (e_1 \cdots e_r)$ gives the principal-submatrix trace. This proof works for indefinite A as well as for covariance matrices.

Exercise 12.14 (Determinant bounds for compressed matrices). Now assume $A \succ 0$. Find the maximum and minimum of $\det(Q^\top A Q)$ under $Q^\top Q = I_r$, and deduce bounds for the determinant of the leading $r \times r$ principal submatrix.

Solution. The compressed matrix is positive definite, since $y^\top Q^\top A Q y > 0$ for $y \neq 0$. Its eigenvalues satisfy the positive bounds $0 < \lambda_{n-r+k} \leq \mu_k \leq \lambda_k$. Multiplication, which preserves inequalities between positive numbers, gives

$$\prod_{k=1}^r \lambda_{n-r+k} \leq \det(Q^\top A Q) \leq \prod_{k=1}^r \lambda_k.$$

Both bounds are attained by bottom or top eigenvectors, respectively. The principal-submatrix result follows by choosing the first r coordinate vectors as columns of Q .

Exercise 12.15 (A diagonal-product consequence). For $A = (a_{ij}) \succ 0$, prove $\prod_{k=1}^r \lambda_{n-r+k} \leq \prod_{k=1}^r a_{kk}$ for $1 \leq r \leq n$.

Solution. Let B be the leading principal submatrix. The preceding determinant bound gives $\prod_{k=1}^r \lambda_{n-r+k} \leq \det B$. Write $B = R^\top R$ by applying Gram–Schmidt to a square root of B , with R upper triangular. Then $\det B = \prod_{j=1}^r r_{jj}^2$, whereas $b_{jj} = \sum_{i \leq j} r_{ij}^2 \geq r_{jj}^2$. Thus $\det B \leq \prod_j b_{jj} = \prod_{j=1}^r a_{jj}$. Combining the two inequalities proves the claim, including $r = n$.

SVD structure and PCA calculations

Exercise 12.16 (Constructing the right singular directions). Let $A \in \mathbb{R}^{m \times n}$ have rank r , and let $P = (P_1 \ P_2)$ be orthogonal with $P^\top A A^\top P = \text{diag}(D^2, 0)$, where D is an invertible diagonal $r \times r$ matrix. Put $Q_1 = A^\top P_1 D^{-1}$ and let $Q_2 \in \mathbb{R}^{n \times (n-r)}$ satisfy $Q_1^\top Q_2 = 0$. Prove $P^\top A (Q_1 \ Q_2) = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$.

Solution. $Q_1^\top Q_1 = D^{-1} P_1^\top A A^\top P_1 D^{-1} = I_r$. Also $P_2^\top A A^\top P_2 = 0$, so $\|A^\top P_2\|_F^2 = 0$ and $P_2^\top A = 0$. The relation $A A^\top P_1 = P_1 D^2$ gives $A Q_1 = P_1 D$. In particular $P_1^\top A Q_1 = D$. Since $Q_1^\top = D^{-1} P_1^\top A$, the assumed $Q_1^\top Q_2 = 0$ gives $P_1^\top A Q_2 = 0$. The other two blocks vanish by $P_2^\top A = 0$. The requested block identity follows; it does not require Q_2 to be orthonormal. To obtain an orthogonal SVD matrix Q , choose Q_2 to be an orthonormal basis of $\ker Q_1^\top$.

Exercise 12.17 (Grouping equal singular values). Suppose $A = \sum_{i=1}^r \sigma_i u_i v_i^\top$ is an SVD. For each distinct positive singular value a_j , let $W_j = \sum_{\{i:\sigma_i=a_j\}} u_i v_i^\top$. Prove $W_j W_j^\top W_j = W_j$, and for $j \neq l$, $W_l^\top W_j = 0$ and $W_l W_j^\top = 0$.

Solution. Let I_j be the index set in the definition. Orthonormality gives

$$W_j^\top W_j = \sum_{i \in I_j} v_i v_i^\top, \quad W_j W_j^\top = \sum_{i \in I_j} u_i u_i^\top.$$

Multiplying the second equality by W_j makes all cross terms zero and keeps $u_i v_i^\top$ for each $i \in I_j$, giving W_j . For distinct groups the index sets are disjoint. In $W_l^\top W_j$, every summand contains $u_h^\top u_i = 0$ with $h \in I_l, i \in I_j$; in $W_l W_j^\top$, every summand instead contains $v_h^\top v_i = 0$. Thus both products vanish.

Exercise 12.18 (Range and kernel in an SVD). For $A = P_1 D Q_1^\top$ with D invertible and $P = (P_1 \ P_2)$, $Q = (Q_1 \ Q_2)$ orthogonal, prove $\text{col}(A) = \text{col}(P_1)$ and $\ker A = \text{col}(Q_2)$.

Solution. Every output of A is in $\text{col}(P_1)$, while $A Q_1 D^{-1} = P_1$, giving the reverse inclusion. Also $Ax = 0$ holds exactly when $Q_1^\top x = 0$, because P_1 has independent columns and D is invertible. The orthogonal expansion $x = Q_1 Q_1^\top x + Q_2 Q_2^\top x$ then shows that this kernel is exactly $\text{col}(Q_2)$. Thus discarded zero singular directions describe a null space, not missing positive-variance directions.

Exercise 12.19 (Two-variable PCA from observations). For $D = \begin{pmatrix} 12 & 21 \\ 11 & 22 \\ 9 & 18 \\ 8 & 19 \end{pmatrix}$, compute the mean,

centered matrix, covariance, principal directions, score columns and explained-variance proportions. Reconstruct the observations with one component and compute the total squared reconstruction error.

Solution. The mean is $(10, 20)^\top$ and $Z = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{pmatrix}$. Thus $S = \frac{1}{3} \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}$. The sum and

difference directions are $v_1 = (1, 1)^\top / \sqrt{2}$, $v_2 = (1, -1)^\top / \sqrt{2}$, with eigenvalues $6, 2/3$. Their scores are $(3, 3, -3, -3)^\top / \sqrt{2}$ and $(1, -1, 1, -1)^\top / \sqrt{2}$, respectively. The proportions are $9/10$ and $1/10$. Keeping the first component gives

$$D_1 = \begin{pmatrix} 23/2 & 43/2 \\ 23/2 & 43/2 \\ 17/2 & 37/2 \\ 17/2 & 37/2 \end{pmatrix}.$$

Subtracting from D gives residual rows $(1/2, -1/2), (-1/2, 1/2), (1/2, -1/2), (-1/2, 1/2)$. Their squared lengths sum to 2, equal to $(n-1)\lambda_2 = 3(2/3)$. The best-rank theorem proves that no other rank-one centered fit has smaller squared error.

Exercise 12.20 (More variables than observations). For $D = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{pmatrix}$, compute PCA using the observation Gram matrix, then recover the loading and scores. State the covariance rank and the number of nonzero components.

Solution. Put $a = (1, 2, 3, 4)^\top$ and $u = (-1, 0, 1)^\top$. The mean is $2a$ and $Z = ua^\top$, where $a^\top a = 30$, $u^\top u = 2$. Thus $G = ZZ^\top/2 = 15uu^\top$ has unit eigenvector $u/\sqrt{2}$ with eigenvalue 30, and has two zero eigenvalues. The recovered loading is

$$v = \frac{Z^\top(u/\sqrt{2})}{\sqrt{2 \cdot 30}} = \frac{a\sqrt{2}}{\sqrt{60}} = \frac{a}{\sqrt{30}}.$$

Its scores are $Zv = \sqrt{30}u$. The covariance is $S = aa^\top$, of rank one with eigenvalues 30, 0, 0, 0. One component explains all variation and reconstructs Z exactly. The other loadings may be any orthonormal basis of $a^\perp$ and have identically zero scores.

Exercise 12.21 (Covariance and correlation PCA). For the covariance matrix $S = \begin{pmatrix} 9 & 3 \\ 3 & 1 \end{pmatrix}$, compute covariance PCA and correlation PCA. Express the first standardized score as a linear combination of the original centered variables.

Solution. $S = aa^\top$ with $a = (3, 1)^\top$. Covariance PCA has eigenvalues 10, 0 and first loading $(3, 1)^\top/\sqrt{10}$. The standard-deviation matrix is $D_s = \text{diag}(3, 1)$. Thus $R = D_s^{-1}SD_s^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, with eigenvalues 2, 0 and first loading $(1, 1)^\top/\sqrt{2}$ in standardized coordinates. For a centered observation (z_1, z_2) , its standardized score is $(z_1/3 + z_2)/\sqrt{2}$. Both methods retain all variation with one component, but their unit loading directions and score scales are different. Standardization is valid because both marginal variances are positive, despite singularity of S .

Exercise 12.22 (A tie at the cutoff). For $S = \text{diag}(4, 4, 1)$, describe all optimal one-dimensional and two-dimensional PCA subspaces and their explained-variance proportions. For $Z = \text{diag}(2, 2, 1)$, describe all best rank-one Frobenius approximations and determine the minimum squared error.

Solution. The total variance is 9. Every line in $\text{span}(e_1, e_2)$ is an optimal first direction, explaining 4/9. The optimal two-dimensional space is uniquely $\text{span}(e_1, e_2)$, explaining 8/9, because the gap after the second eigenvalue is positive. An orthonormal basis within that plane can still be rotated arbitrarily. For the separate approximation problem, $Z^\top Z = S$. The low-rank proof shows that an optimal rank-one matrix must equal ZP , where $P = vv^\top$ projects onto an optimal line with $v = (a, b, 0)^\top$ and $a^2 + b^2 = 1$. Thus every best approximation is $2vv^\top$ and has squared error $4 + 1 = 5$. Conversely each such choice attains that error. A rank-zero competitor has error 9, so no additional zero-rank minimizer is missing.

Exercise 12.23 (The affine reconstruction problem). For observations $d_1, \dots, d_n \in \mathbb{R}^p$, fix a direction subspace W with projector P . Prove that an affine translate $a + W$ minimizes the sum of squared distances for this direction precisely when it contains the observation mean. Then identify the best direction space of dimension k in terms of the sample covariance.

Solution. Write $d_i - a = (d_i - \bar{d}) + (\bar{d} - a)$. Since $\sum_i (d_i - \bar{d}) = 0$, expansion gives

$$\sum_i \|(I - P)(d_i - a)\|^2 = \sum_i \|(I - P)(d_i - \bar{d})\|^2 + n \|(I - P)(\bar{d} - a)\|^2.$$

The first term does not depend on a . The second is zero exactly when $\bar{d} - a \in W$, equivalently $\bar{d} \in a + W$. For such a translate, the objective is $(n - 1)[\text{tr } S - \text{tr}(PS)]$. The trace variational theorem makes it minimal for a span of the top k eigenvectors, with the cutoff-tie choices described there. The minimum is $(n - 1) \sum_{i > k} \lambda_i(S)$.

Exercise 12.24 (Equicorrelation interpreted as PCA). For the $p \times p$ equicorrelation covariance with $p \geq 2$ and $-1/(p-1) < t < 1$, determine the leading PCA subspace and its variance proportion when $t > 0$, $t = 0$, and $t < 0$. What happens at the two positive-semidefinite endpoints?

Solution. The total variance is p . The constant direction has eigenvalue $a = 1 + (p-1)t$, and its perpendicular complement has eigenvalue $b = 1 - t$. If $t > 0$, $a > b$ and the leading subspace is uniquely $\mathbb{R}\mathbf{1}$, explaining a/p . If $t = 0$, all eigenvalues equal 1, so every line is optimal and explains $1/p$. If $t < 0$, $b > a$, and any line in $\mathbf{1}^\perp$ is optimal, explaining b/p ; the entire leading eigenspace has dimension $p-1$. At $t = 1$, $a = p, b = 0$, so one constant-direction component captures everything. At $t = -1/(p-1)$, $a = 0, b = p/(p-1)$, and the $p-1$ perpendicular directions together capture all variance, each contributing $1/(p-1)$. For $p = 2$ the perpendicular eigenspace is one-dimensional.

Further matrix inequalities

In this supplement all matrices are real. For symmetric matrices, $A \geq B$ means $A - B$ is positive semidefinite, and $A > B$ means $A - B$ is positive definite. Orders are positive integers. $\det$ denotes determinant; $\|\cdot\|_F$ denotes the Frobenius norm.

Schwarz, trace, and rank

Exercise 12.25. Prove $(a^\top b)^2 \leq (a^\top a)(b^\top b)$, with equality exactly when a, b are dependent, by each of these methods: (a) use $ab^\top - ba^\top$; (b) for $b \neq 0$, use $I - bb^\top/(b^\top b)$; (c) use the positive semidefinite matrices $c_i c_i^\top$, where $c_i = (a_i, b_i)^\top$.

Solution. (a) Expanding the sum of squared entries gives

$$\|ab^\top - ba^\top\|_F^2 = \sum_{i,j} (a_i b_j - b_i a_j)^2 = 2(\|a\|^2 \|b\|^2 - (a^\top b)^2) \geq 0.$$

Equality means $ab^\top = ba^\top$. If $b \neq 0$, multiply by b to get $a\|b\|^2 = b(a^\top b)$, so the vectors are dependent; if $b = 0$, dependence and equality both hold. The converse is immediate. (b) The indicated matrix P is symmetric idempotent, so $a^\top P a = \|Pa\|^2 \geq 0$. Expanding gives the inequality, and equality means $Pa = 0$, exactly $a \in \text{span}(b)$. (c) The sum is $\sum_i c_i c_i^\top = \begin{pmatrix} \|a\|^2 & a^\top b \\ a^\top b & \|b\|^2 \end{pmatrix}$, a positive semidefinite Gram matrix, whose determinant is nonnegative. It is singular exactly when the two columns of $(a \mid b)$ are dependent: $\ker(C^\top C) = \ker C$ proves this precisely. Thus equality is again equivalent to dependence.

Exercise 12.26. Let $A = (a_{ij})$ be positive semidefinite. Prove $(x^\top A y)^2 \leq (x^\top A x)(y^\top A y)$, with equality exactly when Ax, Ay are dependent. Deduce $|a_{ij}| \leq \max_k a_{kk}$.

Solution. Apply Schwarz to $A^{1/2}x, A^{1/2}y$. Equality means that a nontrivial combination $z = ax + by$ belongs to $\ker A^{1/2} = \ker A$, exactly the dependence of Ax, Ay . With $x = e_i, y = e_j$, obtain $|a_{ij}| \leq \sqrt{a_{ii}a_{jj}}$. The diagonal entries are nonnegative, and their geometric mean is no larger than their maximum, proving the result.

Exercise 12.27. For positive definite A , prove $(x^\top y)^2 \leq (x^\top A^{-1}x)(y^\top Ay)$, with equality exactly when $A^{-1}x, y$ are dependent. For fixed $x \neq 0$, show

$$\frac{1}{x^\top A^{-1}x} = \min_{y: y^\top x \neq 0} \frac{y^\top Ay}{(y^\top x)^2}.$$

Deduce, for positive definite A, B ,

$$x^\top (A + B)^{-1}x \leq \frac{(x^\top A^{-1}x)(x^\top B^{-1}x)}{x^\top (A^{-1} + B^{-1})x}.$$

Solution. Schwarz applied to $A^{-1/2}x, A^{1/2}y$ gives the first inequality; its equality condition becomes dependence after multiplication by $A^{-1/2}$. Dividing by the positive quantities gives the proposed lower bound, attained at $y = A^{-1}x$, for which $y^\top x > 0$. Let $\psi(A) = 1/(x^\top A^{-1}x)$. For each admissible y ,

$$\frac{y^\top (A + B)y}{(y^\top x)^2} \geq \psi(A) + \psi(B).$$

Taking the minimum gives $\psi(A + B) \geq \psi(A) + \psi(B)$. All terms are positive, so taking reciprocals and combining the two fractions proves the stated bound.

Exercise 12.28. (a) Prove $(\sum_i x_i)^2 \leq n \sum_i x_i^2$, with equality exactly when all x_i agree. (b) If all eigenvalues of a real square A are real, prove

$$|\operatorname{tr} A/n| \leq \sqrt{\operatorname{tr}(A^2)/n},$$

with equality exactly when all eigenvalues agree. (c) If $A \neq 0$ is symmetric, show $\operatorname{rank} A \geq (\operatorname{tr} A)^2 / \operatorname{tr}(A^2)$.

Solution. (a) Apply Schwarz to (x_i) and the all-ones vector; dependence means all coordinates agree. (b) The trace identities $\operatorname{tr} A = \sum_i \lambda_i$ and $\operatorname{tr}(A^2) = \sum_i \lambda_i^2$ reduce the claim to (a). For completeness these identities require no diagonalizability: choose one real eigenvector as the first basis vector, obtaining a block upper triangular matrix $\begin{pmatrix} \lambda & * \\ 0 & C \end{pmatrix}$. Its characteristic polynomial factors, and its trace and squared trace contributions are $\lambda + \operatorname{tr} C$ and $\lambda^2 + \operatorname{tr}(C^2)$. Induction proves both identities because the remaining eigenvalues stay real. (c) Apply (a) to the $r = \operatorname{rank} A$ nonzero eigenvalues: $(\operatorname{tr} A)^2 \leq r \operatorname{tr}(A^2)$. The denominator is positive because $A \neq 0$ is symmetric.

Exercise 12.29. For real matrices A, B of the same size, prove

- (a) $(\operatorname{tr}(A^\top B))^2 \leq \operatorname{tr}(A^\top A) \operatorname{tr}(B^\top B)$, with equality exactly when one matrix is a scalar multiple of the other;
- (b) $\operatorname{tr}((A^\top B)^2) \leq \operatorname{tr}(A^\top A B^\top B)$, with equality exactly when $AB^\top$ is symmetric;
- (c) $\operatorname{tr}((A^\top B)^2) \leq \operatorname{tr}(AA^\top BB^\top)$, with equality exactly when $A^\top B$ is symmetric.

Solution. (a) This is Schwarz for the Frobenius inner product; viewing all entries as one vector gives the stated equality condition, including zero matrices. For any real square M , direct expansion gives

$$\|M - M^\top\|_F^2 = 2(\operatorname{tr}(M^\top M) - \operatorname{tr}(M^2)).$$

(b) Apply this to $M = AB^\top$. Cyclicity gives $\operatorname{tr}(M^2) = \operatorname{tr}((A^\top B)^2)$ (also use invariance under transpose), and $\operatorname{tr}(M^\top M) = \operatorname{tr}(A^\top A B^\top B)$. The nonnegative difference vanishes exactly when $M = M^\top$. (c) Apply the same identity to $M = A^\top B$, for which $\operatorname{tr}(M^\top M) = \operatorname{tr}(AA^\top BB^\top)$.

Exercise 12.30. For any real square A , prove $\operatorname{tr}(A^2) \leq \operatorname{tr}(A^\top A)$, with equality exactly when A is symmetric.

Solution. Expanding the Frobenius square yields $\|A - A^\top\|_F^2 = 2 \operatorname{tr}(A^\top A) - 2 \operatorname{tr}(A^2)$. It is nonnegative and is zero exactly when $A = A^\top$.

Determinants and positive order

Exercise 12.31. For positive semidefinite $n \times n$ matrices A, B , prove $\det(A + B) \geq \det A + \det B$. Determine every equality case.

Solution. If both are singular, the right side is zero and the inequality follows from positive semidefiniteness of their sum. Equality then means the sum is singular. Otherwise suppose A is positive definite, exchanging the names if needed. Put $C = A^{-1/2}BA^{-1/2}$, with eigenvalues $\lambda_i \geq 0$. Dividing the proposed inequality by $\det A > 0$ reduces it to $\prod_i(1 + \lambda_i) \geq 1 + \prod_i \lambda_i$, which follows by expanding the product into nonnegative terms. For $n = 1$ there is always equality. For $n \geq 2$, the omitted terms include $\sum_i \lambda_i$, so equality in this nonsingular case requires every $\lambda_i = 0$, or $B = 0$. Thus all equality cases are: $n = 1$, or $A + B$ singular, or $A = 0$, or $B = 0$.

Exercise 12.32. If $A \geq B \geq 0$, show $\det A \geq \det B$. Prove that equality holds exactly when both matrices are singular, or when both are nonsingular and $A = B$.

Solution. If B is singular, its determinant is zero and equality holds exactly when A is singular too. If B is positive definite, write $A = B^{1/2}(I + C)B^{1/2}$, where $C = B^{-1/2}(A - B)B^{-1/2} \geq 0$. Then $\det A / \det B = \prod_i(1 + \lambda_i(C)) \geq 1$, with equality precisely when all eigenvalues of C vanish, or $A = B$.

Exercise 12.33. If $0 \leq A \leq I_n$, prove $A = I_n \iff \det A = 1$.

Solution. The eigenvalues lie in $[0, 1]$, since for a unit eigenvector v , $0 \leq v^\top Av \leq 1$. Their product is one exactly when every eigenvalue is one. Orthogonal diagonalization then gives $A = I_n$. The converse is immediate.

Exercise 12.34. Prove the arithmetic–geometric mean inequality $(\prod_i \lambda_i)^{1/n} \leq \sum_i \lambda_i / n$ for nonnegative λ_i , with equality exactly when all are equal. Deduce $(\det A)^{1/n} \leq \operatorname{tr} A / n$ for positive semidefinite A , and prove that equality holds exactly when $A = \alpha I_n$, $\alpha \geq 0$.

Solution. For $t > 0$, the function $t - 1 - \log t$ has derivative $1 - 1/t$, so its unique minimum is zero at $t = 1$. If every $\lambda_i > 0$, let $a = \sum_i \lambda_i / n$. Sum $\log(\lambda_i/a) \leq \lambda_i/a - 1$ to obtain $\log(\prod_i \lambda_i / a^n) \leq 0$. Equality requires every $\lambda_i = a$. If some $\lambda_i = 0$, the product is zero, and equality requires that their nonnegative sum be zero, so all are zero. Apply this scalar result to the eigenvalues of A . Equality makes its orthogonal diagonal form αI_n , and hence $A = \alpha I_n$.

Exercise 12.35. For positive semidefinite A and positive definite X with $\det X = 1$, prove $\operatorname{tr}(AX) / n \geq (\det A)^{1/n}$. Show that equality holds exactly when $A = 0$, or when A is positive definite and $X = (\det A)^{1/n} A^{-1}$.

Solution. Apply the preceding arithmetic–geometric mean inequality to $C = X^{1/2}AX^{1/2}$. Its trace is $\operatorname{tr}(AX)$, and its determinant is $\det A$. Equality means $C = \alpha I_n$. If $\alpha = 0$, invertibility of $X^{1/2}$ gives $A = 0$. Otherwise $A = \alpha X^{-1} > 0$; taking determinants yields $\alpha = (\det A)^{1/n}$, hence the stated X . Each of these choices attains equality by substitution.

Exercise 12.36. For nonzero positive semidefinite A, B , prove

$$(\det(A + B))^{1/n} \geq (\det A)^{1/n} + (\det B)^{1/n}.$$

Determine every equality case using the preceding trace representation.

Solution. If $S = A + B$ is singular, a nonzero $z \in \ker S$ satisfies $z^\top A z + z^\top B z = 0$, so both terms vanish and both A, B are singular. Equality is then zero equals zero. If $S > 0$, choose $X = (\det S)^{1/n} S^{-1}$, with $\det X = 1$. The preceding exercise gives

$$(\det S)^{1/n} = \operatorname{tr}(SX)/n = \operatorname{tr}(AX)/n + \operatorname{tr}(BX)/n \geq (\det A)^{1/n} + (\det B)^{1/n}.$$

Because neither summand matrix is zero, equality requires both to be positive definite and the same X to be their equality choice. Thus $(\det A)^{1/n} A^{-1} = (\det B)^{1/n} B^{-1}$, equivalent to $A = cB$ for some $c > 0$. Such proportional matrices attain equality. Altogether equality holds exactly when $A + B$ is singular or $A = cB$ with $c > 0$.

Exercise 12.37. For positive semidefinite A, B , prove $0 \leq \operatorname{tr}(AB) \leq \operatorname{tr} A \operatorname{tr} B$ and

$$\sqrt{\operatorname{tr}(AB)} \leq \frac{1}{2}(\operatorname{tr} A + \operatorname{tr} B).$$

Determine every equality case in the second inequality.

Solution. Diagonalize A orthogonally to $\operatorname{diag}(\lambda_i)$, and write C for B in the same coordinates. Then $\operatorname{tr}(AB) = \sum_i \lambda_i c_{ii} \geq 0$, and

$$\operatorname{tr} A \operatorname{tr} B - \operatorname{tr}(AB) = \sum_{i \neq j} \lambda_i c_{jj} \geq 0.$$

Also $\sqrt{ab} \leq (a + b)/2$ for nonnegative a, b , since $(\sqrt{a} - \sqrt{b})^2 \geq 0$. Equality requires equal traces and all the displayed cross terms zero. If either matrix is zero, equal traces force both to be zero. Otherwise choose $\lambda_k > 0$. Then $c_{jj} = 0$ for $j \neq k$; the PSD entry bound forces the corresponding rows and columns of C to vanish. As $C \neq 0$, $c_{kk} > 0$, forcing $\lambda_i = 0$ for $i \neq k$. Equal traces now give $\lambda_k = c_{kk}$. Thus $A = B = aa^\top$ for some nonzero vector a ; taking $a = 0$ includes the zero case. These and only these choices attain equality.

Exercise 12.38. For real $m \times n$ matrices A, B , prove

$$(\det(A^\top B))^2 \leq \det(A^\top A) \det(B^\top B).$$

Prove equality holds exactly when one Gram matrix is singular, or $B = AQ$ for an invertible $n \times n$ matrix Q .

Solution. If a Gram matrix is singular, its corresponding matrix has rank below n , so $A^\top B$ is singular and both sides vanish. Otherwise both have independent columns. Let $P = A(A^\top A)^{-1}A^\top$, the orthogonal projector onto $\operatorname{im} A$. Then $0 \leq B^\top P B \leq B^\top B$, and determinant monotonicity gives

$$\frac{(\det(A^\top B))^2}{\det(A^\top A)} = \det(B^\top P B) \leq \det(B^\top B).$$

Since the upper matrix is positive definite, equality requires $B^\top(I - P)B = 0$. This is the Gram matrix of $(I - P)B$, so equality means $PB = B$. Thus $B = AQ$; its independent columns force Q to be invertible. Conversely this relation gives equality by determinant multiplication.

Inverse and power inequalities

Exercise 12.39. For positive definite A, B , prove $A > B \iff B^{-1} > A^{-1}$, where $>$ means that the difference is positive definite.

Solution. If $A > B$, then $C = B^{-1/2}AB^{-1/2} > I$. Every eigenvalue of C is greater than one, so $I - C^{-1} > 0$. Congruence by $B^{-1/2}$ gives $B^{-1} - A^{-1} > 0$. Apply the implication just proved to B^{-1}, A^{-1} to obtain the reverse direction.

Exercise 12.40. (a) For $a \leq t \leq b$, prove $t^2 - (a+b)t + ab \leq 0$. (b) Let $A > 0$ have order $n \geq 2$ and extreme eigenvalues $M \geq m > 0$. Prove $(M+m)I - A - MmA^{-1} \geq 0$, of rank at most $n-2$. (c) Using the trace-square inequality and (b), prove for every unit x

$$1 \leq (x^\top Ax)(x^\top A^{-1}x) \leq \frac{(M+m)^2}{4Mm}.$$

Solution. (a) Factor the polynomial as $(t-a)(t-b) \leq 0$. (b) In an eigenbasis, the displayed matrix has entries $M+m-\lambda-Mm/\lambda \geq 0$ by (a), with zero at each extreme eigenvalue. If $M > m$, these give at least two zero entries; if $M = m$, all entries vanish. The rank bound follows in both cases. (c) Put $D = A^{1/2}xx^\top A^{-1/2}$. Then $\text{tr}(D^2) = 1$ and $\text{tr}(D^\top D) = (x^\top Ax)(x^\top A^{-1}x)$. The trace-square inequality proves the lower bound. Set $u = x^\top Ax$, $v = x^\top A^{-1}x$. Part (b) gives $u + Mmv \leq M + m$, whence

$$Mmuv \leq (M+m)u - u^2 = \frac{(M+m)^2}{4} - \left(u - \frac{M+m}{2}\right)^2 \leq \frac{(M+m)^2}{4}.$$

Division by $Mm > 0$ finishes the proof. For order one the two bounds are both one, although the stated rank bound would need separate wording.

Exercise 12.41. Suppose $A \geq B \geq 0$. (a) Prove $A^{1/2} \geq B^{1/2}$. (b) Show that $A^2 \geq B^2$ need not hold. (c) Prove that if $AB = BA$, then $A^k \geq B^k$ for every integer $k \geq 2$.

Solution. (a) Let $C = A^{1/2} - B^{1/2}$, symmetric. Expanding the square gives $A - B = A^{1/2}C + CA^{1/2} - C^2$. If $Cx = \lambda x$, $x \neq 0$, and $\lambda < 0$, then

$$x^\top (A - B)x = 2\lambda x^\top A^{1/2}x - \lambda^2 \|x\|^2 < 0,$$

contradicting $A - B \geq 0$. Thus every eigenvalue of C is nonnegative, proving the claim, including singular matrices. (b) Take $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $B = \text{diag}(1, 0)$. The difference $A - B$ is the all-ones matrix

and is positive semidefinite; A, B are positive semidefinite. But $A^2 - B^2 = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}$ has value -2 on $(2, -3)^\top$, so it is not positive semidefinite. (c) Each eigenspace of A is invariant under B , because they commute. Diagonalize the symmetric restriction of B within each such eigenspace to obtain one orthonormal basis diagonalizing both. Their diagonal entries satisfy $\lambda_i \geq \mu_i \geq 0$ by the assumed order. Then $\lambda_i^k \geq \mu_i^k$ proves the matrix inequality after changing coordinates back.

Logarithms and determinant concavity

Exercise 12.42. For positive definite $n \times n$ A , prove $\log \det A \leq \text{tr } A - n$, with equality exactly when $A = I_n$.

Solution. For each positive eigenvalue λ_i , the scalar inequality $\log \lambda_i \leq \lambda_i - 1$ was proved by differentiation in the arithmetic-geometric mean exercise. Summing gives the stated bound. Equality requires every $\lambda_i = 1$; orthogonal diagonalization then gives $A = I_n$, which also attains equality.

Exercise 12.43. (a) For $t > 0$, $0 < \alpha < 1$, prove $t^\alpha \leq \alpha t + 1 - \alpha$, with equality exactly at $t = 1$. (b) For positive semidefinite A, B of the same size, prove

$$(\det A)^\alpha (\det B)^{1-\alpha} \leq \det(\alpha A + (1-\alpha)B).$$

(c) Determine every equality case.

Solution. (a) The derivative of $\alpha t + 1 - \alpha - t^\alpha$ is $\alpha(1 - t^{\alpha-1})$, negative before one and positive after one. Its unique minimum is zero at one. (b) If either matrix is singular, the left side is zero and the right side nonnegative. If both are positive definite, set $C = B^{-1/2}AB^{-1/2}$, with positive eigenvalues λ_i . Divide both sides by $\det B$; the claim becomes $\prod_i \lambda_i^\alpha \leq \prod_i (\alpha \lambda_i + 1 - \alpha)$, which follows from (a). (c) In the positive definite case, equality requires every $\lambda_i = 1$, or $A = B$. In the singular case equality means the mixture is singular. For positive weights, its kernel is $\ker A \cap \ker B$, since a zero quadratic value in the sum forces both nonnegative terms to vanish. Thus equality holds exactly when $A = B$, or the two matrices have a common nonzero kernel vector.

Exercise 12.44. For positive definite matrices $A_1, \dots, A_k$ and positive weights α_i summing to one, prove

$$\prod_{i=1}^k (\det A_i)^{\alpha_i} \leq \det \left(\sum_{i=1}^k \alpha_i A_i \right).$$

Determine when equality holds.

Solution. Induct on k . For $k = 1$ equality is automatic. For $k > 1$, let $s = \sum_{i < k} \alpha_i$ and $C = \sum_{i < k} (\alpha_i/s) A_i$. The induction hypothesis bounds the first $k-1$ determinant factors by $(\det C)^s$; the two-matrix inequality then gives $(\det C)^s (\det A_k)^{1-s} \leq \det(sC + (1-s)A_k)$. All intermediate matrices are positive definite. Equality requires $A_1 = \dots = A_{k-1}$ by induction and $C = A_k$ by the two-matrix equality case. Thus equality holds precisely when all A_i agree.

Bordered matrices and volume bounds

Exercise 12.45. Let $A > 0$ and $Z = \begin{pmatrix} A & b \\ b^\top & \alpha \end{pmatrix}$. Prove (a) $\det Z \leq \alpha \det A$, with equality exactly when $b = 0$; (b) $Z > 0 \iff \det Z > 0$.

Solution. With $L = \begin{pmatrix} I & -A^{-1}b \\ 0 & 1 \end{pmatrix}$, multiplication gives $L^\top Z L = \text{diag}(A, \alpha - b^\top A^{-1}b)$, and $\det L = 1$. Hence $\det Z = \det A(\alpha - b^\top A^{-1}b)$. Since $A^{-1} > 0$, the subtracted term is nonnegative and vanishes exactly for $b = 0$, proving (a). Congruence by invertible L preserves positive definiteness, so $Z > 0$ exactly when $\alpha - b^\top A^{-1}b > 0$. As $\det A > 0$, this is exactly $\det Z > 0$.

Exercise 12.46. Let $A \geq 0$ and $Z = \begin{pmatrix} A & b \\ b^\top & \alpha \end{pmatrix}$. Prove $Z \geq 0$ exactly when $\alpha \geq 0$ and $\alpha A - bb^\top \geq 0$. Obtain the positive definite bordered criterion and determinant bound of the preceding exercise as a special case.

Solution. If $Z \geq 0$, then $\alpha \geq 0$ by its last diagonal entry. If $\alpha > 0$, complete the square:

$$\begin{pmatrix} x \\ t \end{pmatrix}^\top Z \begin{pmatrix} x \\ t \end{pmatrix} = \alpha \left(t + \frac{b^\top x}{\alpha} \right)^2 + x^\top \left(A - \frac{bb^\top}{\alpha} \right) x.$$

This proves both directions in this case. If $\alpha = 0$ and $b \neq 0$, choose $x = b$ and take t sufficiently negative to make the quadratic expression negative. Thus $Z \geq 0$ forces $b = 0$; conversely $b = 0$ makes $Z = \text{diag}(A, 0) \geq 0$. The condition $-bb^\top \geq 0$ also forces $b = 0$, proving the claimed equivalence at zero. For $A > 0$, completing the square in x instead gives $(x + tA^{-1}b)^\top A(x + tA^{-1}b) + (\alpha - b^\top A^{-1}b)t^2$. It is positive definite exactly when the last coefficient is positive. The same change of coordinates has determinant one, giving $\det Z = \det A(\alpha - b^\top A^{-1}b) \leq \alpha \det A$, with equality exactly at $b = 0$, as required.

Exercise 12.47. For $A > 0$, show that $Z = \begin{pmatrix} A & b \\ b^\top & b^\top A^{-1}b \end{pmatrix}$ is positive semidefinite and singular. Find a zero-eigenvalue eigenvector.

Solution. Its quadratic form is $(x + tA^{-1}b)^\top A(x + tA^{-1}b) \geq 0$. The nonzero vector $(-A^{-1}b, 1)^\top$ is annihilated by both block rows: the first gives $-b + b = 0$, the second $-b^\top A^{-1}b + b^\top A^{-1}b = 0$. Thus it is a zero-eigenvalue eigenvector and Z is singular.

Exercise 12.48. (a) For positive definite $A = (a_{ij})$, prove $\det A \leq \prod_i a_{ii}$, with equality exactly when A is diagonal. (b) Deduce for every real square A that $(\det A)^2 \leq \prod_i \sum_j a_{ij}^2$. (c) Determine every equality case in (b).

Solution. (a) Induct on order, the scalar case being equality. Partition $A = \begin{pmatrix} C & b \\ b^\top & d \end{pmatrix} > 0$, so $C > 0$. The bordered determinant identity gives $\det A = \det C(d - b^\top C^{-1}b) \leq d \det C \leq d \prod_{i < n} a_{ii}$. Since all diagonal entries are positive, equality requires both $b = 0$ and equality for C , hence A diagonal. The converse follows directly. (b) For nonsingular A , apply (a) to $AA^\top > 0$, whose diagonal entries are its row squared norms and determinant is $(\det A)^2$. For singular A , the left side is zero and the right side nonnegative. (c) In the nonsingular case, equality means $AA^\top$ diagonal, or mutually orthogonal rows. In the singular case equality means at least one row squared norm is zero, or at least one zero row. Thus equality holds exactly when a row is zero or all rows are mutually orthogonal.

Exercise 12.49. (a) For real symmetric matrices, show that $\det A = \prod_i a_{ii} \iff A$ is diagonal remains true in order two, but can fail in every order at least three. (b) Prove that a real symmetric matrix is diagonal exactly when its multiset of eigenvalues equals its multiset of diagonal entries.

Solution. (a) For $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, the equality is $ad - b^2 = ad$, equivalent to $b = 0$. In order three, $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is nondiagonal but both determinant and diagonal product vanish. Appending an identity block gives the same counterexample in every larger order. (b) If the multisets agree, then

$$\sum_i a_{ii}^2 = \sum_i \lambda_i^2 = \operatorname{tr}(A^2) = \sum_i a_{ii}^2 + 2 \sum_{i < j} a_{ij}^2.$$

Thus every off-diagonal entry vanishes. Conversely a diagonal matrix has exactly its diagonal entries as its eigenvalues.

Inverse and trace comparisons

Exercise 12.50. (a) If $A > 0$, show $A + A^{-1} - 2I \geq 0$. (b) When is this matrix positive definite? (c) For positive definite A, B , prove $\operatorname{tr}((A^{-1} - B^{-1})(A - B)) \leq 0$.

Solution. (a,b) In an eigenbasis of A , the entries are $\lambda + \lambda^{-1} - 2 = (\lambda - 1)^2/\lambda \geq 0$. They are all positive exactly when 1 is not an eigenvalue of A . (c) Put $C = A^{-1/2}BA^{-1/2} > 0$. Cyclicity of trace gives

$$\operatorname{tr}((A^{-1} - B^{-1})(A - B)) = 2n - \operatorname{tr}(A^{-1}B) - \operatorname{tr}(B^{-1}A) = -\operatorname{tr}(C + C^{-1} - 2I) \leq 0$$

by part (a).

Exercise 12.51. If A and $A + B$ are positive definite, prove

$$\frac{\det(A + B)}{\det A} \leq \exp(\operatorname{tr}(A^{-1}B)),$$

with equality exactly when $B = 0$.

Solution. Set $C = A^{-1/2}(A + B)A^{-1/2} > 0$. The log-determinant inequality gives

$$\log \frac{\det(A + B)}{\det A} = \log \det C \leq \operatorname{tr} C - n = \operatorname{tr}(A^{-1}B).$$

Exponentiation preserves the inequality. Equality is equivalent to $C = I$, equivalent to $B = 0$.

Exercise 12.52. Let $A > 0$, let B be symmetric, and assume $C = A + B$ is nonsingular. Prove $C^{-1}BC^{-1} \leq A^{-1} - C^{-1}$. Show that the inequality is strict exactly when B is nonsingular.

Solution. Multiplying the proposed difference by C on both sides gives

$$C(A^{-1} - C^{-1} - C^{-1}BC^{-1})C = CA^{-1}C - C - B = BA^{-1}B.$$

Thus the difference is $C^{-1}BA^{-1}BC^{-1} = (A^{-1/2}BC^{-1})^\top (A^{-1/2}BC^{-1})$, which is positive semidefinite. It is positive definite exactly when $A^{-1/2}BC^{-1}$ has trivial kernel, exactly when B is nonsingular. No positive-definiteness assumption on C is required.

Block and entrywise products

Exercise 12.53. Define the Hadamard product by $(A \odot B)_{ij} = a_{ij}b_{ij}$. (a) For $\lambda = (\lambda_i)$, $\Lambda = \operatorname{diag}(\lambda_i)$, prove $\lambda^\top (A \odot B) \lambda = \operatorname{tr}(A\Lambda B^\top \Lambda)$. (b) If $A, B > 0$, prove $A \odot B > 0$.

Solution. (a) The trace is $\sum_{i,j} a_{ij}\lambda_j b_{ij}\lambda_i$, exactly the left-hand quadratic expression. (b) For nonzero λ , $D = \Lambda B \Lambda$ is positive semidefinite and nonzero: if $\lambda_i \neq 0$, its diagonal entry $\lambda_i^2 b_{ii} > 0$. Consequently $A^{1/2} D A^{1/2}$ is positive semidefinite and nonzero, so its eigenvalues have positive sum. By (a) and trace cyclicity, $\lambda^\top (A \odot B) \lambda = \operatorname{tr}(AD) = \operatorname{tr}(A^{1/2} D A^{1/2}) > 0$. The entrywise product is symmetric, completing positive definiteness.

Exercise 12.54. Let $A > 0$, $B \in \mathbb{R}^{n \times m}$, and symmetric $X \in \mathbb{R}^{m \times m}$. Prove

$$\begin{pmatrix} A & B \\ B^\top & X \end{pmatrix} \geq 0 \iff X - B^\top A^{-1} B \geq 0,$$

and prove the corresponding equivalence with both inequalities strict.

Solution. For any u, v , complete the square:

$$u^\top A u + 2u^\top B v + v^\top X v = (u + A^{-1} B v)^\top A (u + A^{-1} B v) + v^\top (X - B^\top A^{-1} B) v.$$

If the last matrix is positive semidefinite, this proves the forward quadratic bound. Conversely choose $u = -A^{-1} B v$ to isolate that matrix. For strict positivity, if $v \neq 0$ the second term is positive; if $v = 0, u \neq 0$ the first is positive. Conversely the same substitution with nonzero v forces strict positivity of the last matrix.

Exercise 12.55. Let $Z = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix} > 0$. (a) Prove $\det Z \leq \det A \det D$, with equality exactly at $B = 0$. (b) If B is square, prove $(\det B)^2 < \det A \det D$.

Solution. The diagonal blocks are positive definite, and the preceding criterion makes $S = D - B^\top A^{-1}B > 0$. Congruence by $\begin{pmatrix} I & -A^{-1}B \\ 0 & I \end{pmatrix}$, of determinant one, gives $\det Z = \det A \det S$. Since $0 < S \leq D$, determinant monotonicity gives (a), with equality exactly when $S = D$, or $B^\top A^{-1}B = 0$. The latter is a Gram matrix, and vanishes exactly for $B = 0$. (b) Put $C = B^\top A^{-1}B$, so $0 \leq C < D$. If C is singular, $\det C = 0 < \det D$. Otherwise congruence by $C^{-1/2}$ gives eigenvalues strictly above one, hence again $\det C < \det D$. Now $\det C = (\det B)^2 / \det A$, proving the claim.

Exercise 12.56. For every real rectangular A , prove $\begin{pmatrix} I & A \\ A^\top & A^\top A \end{pmatrix} \geq 0$.

Solution. The matrix factors as $\begin{pmatrix} I \\ A^\top \end{pmatrix} (I \mid A)$. Its quadratic form at (x, y) is therefore $\|x + Ay\|^2 \geq 0$. The factorization also makes symmetry explicit.

Exercise 12.57. For $A, B \in \mathbb{R}^{m \times n}$, prove

- (a) $Z = \begin{pmatrix} I_m + AA^\top & A + B \\ (A + B)^\top & I_n + B^\top B \end{pmatrix} \geq 0$;
- (b) $I_n + B^\top B \geq (A + B)^\top (I_m + AA^\top)^{-1} (A + B)$;
- (c) if $m = n$, then $(\det(A + B))^2 \leq \det(I + B^\top B) \det(I + AA^\top)$.

Solution. (a) Its quadratic form at (x, y) is $\|x + By\|^2 + \|A^\top x + y\|^2$, a sum of nonnegative terms. (b) The first block is positive definite, since $x^\top (I + AA^\top)x = \|x\|^2 + \|A^\top x\|^2 > 0$ for $x \neq 0$. Apply the Schur-complement criterion to obtain the inequality. (c) Both matrices in (b) are positive semidefinite, so determinant monotonicity gives

$$\frac{(\det(A + B))^2}{\det(I + AA^\top)} \leq \det(I + B^\top B).$$

The denominator is positive; multiplying proves the claim.

Exercise 12.58. For positive definite A, B and symmetric X of the same order, prove

$$\begin{pmatrix} A + B & A \\ A & A + X \end{pmatrix} \geq 0 \iff X \geq -(A^{-1} + B^{-1})^{-1}.$$

Solution. The Schur-complement criterion makes the left condition equivalent to $X \geq -A + A(A + B)^{-1}A$. Let $H = A - A(A + B)^{-1}A = A(A + B)^{-1}B$. All three factors in the last expression are nonsingular, and

$$H^{-1} = B^{-1}(A + B)A^{-1} = B^{-1} + A^{-1}.$$

Thus $H = (A^{-1} + B^{-1})^{-1}$, and substitution proves the desired condition. This also establishes symmetry and positive definiteness of H .

Extreme eigenvalues and variation

Exercise 12.59. For real symmetric A with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$, prove the Rayleigh bounds $\lambda_n \leq x^\top Ax / (x^\top x) \leq \lambda_1$ for $x \neq 0$. Show the bounds are attained and express them as a minimum and maximum. Explain why these variational descriptions are useful.

Solution. In an orthonormal eigenbasis, write $x = \sum_i c_i q_i$. Its quotient is $\sum_i \lambda_i c_i^2 / \sum_i c_i^2$, a weighted average of the eigenvalues. This proves both bounds. Taking $x = q_1$, respectively q_n , attains them, so they are the maximum and minimum over nonzero x , equivalently over unit vectors. The representation permits bounds and comparisons of extreme eigenvalues through quadratic forms, without solving a characteristic equation; the next exercise demonstrates this.

Exercise 12.60. For real symmetric A, B , use the Rayleigh representations to prove

$$\lambda_1(A + B) \leq \lambda_1(A) + \lambda_1(B), \quad \lambda_n(A + B) \geq \lambda_n(A) + \lambda_n(B).$$

If $B \geq 0$, show also $\lambda_1(A + B) \geq \lambda_1(A)$ and $\lambda_n(A + B) \geq \lambda_n(A)$.

Solution. For each unit x , $x^\top(A + B)x \leq \lambda_1(A) + \lambda_1(B)$. Taking its maximum gives the first bound. Bounding each summand below and taking the minimum gives the second. When $B \geq 0$, every unit x satisfies $x^\top(A + B)x \geq x^\top Ax$. Taking maxima proves monotonicity of the largest eigenvalue, and taking minima proves monotonicity of the smallest eigenvalue.

Concavity of determinant functions

Exercise 12.61. A real-valued function on positive definite matrices is convex if $f(tA + (1 - t)B) \leq tf(A) + (1 - t)f(B)$ for $0 < t < 1$, and concave if the inequality is reversed. Prove (a) $\log \det X$ is concave; (b) for order $n \geq 2$, $\det X$ is neither convex nor concave; (c) $(\det X)^{1/n}$ is concave for $n \geq 2$.

Solution. (a) The determinant inequality with positive weights gives $\det(tA + (1 - t)B) \geq (\det A)^t (\det B)^{1-t}$. All determinants are positive, so taking logarithms gives precisely concavity of $\log \det$. (b) In order two take $A = I_2$, $B = \text{diag}(\beta, 5)$, $\beta > 0$, and $t = 1/2$. Then

$$\frac{1}{2} \det A + \frac{1}{2} \det B - \det((A + B)/2) = \beta - 1.$$

This difference is negative at $\beta = 1/2$, contradicting convexity, and positive at $\beta = 2$, contradicting concavity. Appending an identity block gives the same counterexamples in every larger order. For order one the determinant is linear, so the dimension restriction is essential. (c) Apply the determinant-root inequality to $tA, (1 - t)B$:

$$(\det(tA + (1 - t)B))^{1/n} \geq t(\det A)^{1/n} + (1 - t)(\det B)^{1/n},$$

since $\det(tA) = t^n \det A$. This is the required concavity.